

The probabilistic Planck gap: $\tau\Delta E \geq (1 - 2\epsilon)^2\hbar^2/(4A)$, and why it is resource-relative

The worst-case theorem of the derivation note has no quantum instance. Its probabilistic replacement does, and it decides the universality question that the worst-case version left open. Let the apparatus confine the particle's position spread to L and its momentum spread to P , and set $A = LP$, the phase-space area available to a preparation; the uncertainty relation forces $A \geq \hbar/2$. Then for **every** protocol of quantum instruments, of any number, at any times, with any intermediate outcomes and any final measurement, distinguishing the falling from the inertial hypothesis with error probability at most ϵ requires

$$\boxed{\frac{F\tau L}{\hbar} + \frac{F\tau^2 P}{2m\hbar} \geq 1 - 2\epsilon, \quad \text{hence} \quad \tau\Delta E = \frac{F^2\tau^3}{2m} \geq \frac{(1 - 2\epsilon)^2\hbar^2}{4A}},$$

the second form after optimizing the apparatus shape at fixed A (Theorem 2, Corollary 3). At the minimal area $A = \hbar/2$ this is $\tau\Delta E \geq (1 - 2\epsilon)^2\hbar/2$, the same combination $F^2\tau^3/m$ that Newton's geometry sends to zero and that the classical theorem bounds by the marks' phase-space cost. A single-shot protocol obeys the sharper condition with $1 - 2\epsilon$ replaced by $\arccos(2\sqrt{\epsilon(1 - \epsilon)})$, which is $\pi/2$ at $\epsilon = 0$ (Theorem 1).

The universality question is decided, and the answer is negative without the resource bound. For every $\tau\Delta E > 0$ there is a preparation and a measurement distinguishing the hypotheses with error probability $O(\tau\Delta E/\hbar)$, built from two narrow packets separated by $d = \pi\hbar/(F\tau)$ (Proposition 4). The separation grows without bound as the force shrinks, saturating Theorem 2, so the theorem is tight in L and the failure of universality is exactly the failure of confinement. Protocol-universality over instruments survives in full; universality over preparations fails. Ozawa's measurements breaking the standard quantum limit for free-mass position belong to the same phenomenon and are covered by the resource-relative statement.

Methods are standard: Mandelstam–Tamm for the single-shot bound, a telescoping hybrid argument for the many-instrument case. The contribution is the protocol-universal statement in the Galileo geometry with explicit constants, and a counterexample that matches it. Exploratory; no ledger promotion.

1. The two hypotheses differ by a phase-space displacement

Transverse coordinate $\hat{y}$, momentum $\hat{p}$, mass $m > 0$, on $[0, \tau]$. Hypothesis I evolves with $H_I = \hat{p}^2/2m$; hypothesis F with $H_F = \hat{p}^2/2m - F\hat{y}$. Write

$$D(a, b) = \exp\left[\frac{i}{\hbar}(b\hat{y} - a\hat{p})\right], \quad D^\dagger\hat{y}D = \hat{y} + a, \quad D^\dagger\hat{p}D = \hat{p} + b.$$

Lemma 1. $W(t) = U_I(t)^\dagger U_F(t) = e^{i\varphi(t)}D(\alpha(t), \beta(t))$ with

$$\alpha(t) = -\frac{Ft^2}{2m}, \quad \beta(t) = Ft,$$

and a real phase φ .

Proof. Under H_F the Heisenberg operators are $\hat{y}(t) = \hat{y} + \hat{p}t/m + Ft^2/2m$ and $\hat{p}(t) = \hat{p} + Ft$, so $U_F(t) = e^{i\varphi}D(Ft^2/2m, Ft)U_I(t)$ for some phase. Free evolution conjugates a displacement to a displacement, $U_I(t)^\dagger D(a, b)U_I(t) = D(a - bt/m, b)$, and $Ft^2/2m - Ft \cdot t/m = -Ft^2/2m$. $\square$

The product of the two components is the programme's combination:

$$|\alpha(\tau)\beta(\tau)| = \frac{F\tau^2}{2m} \cdot F\tau = \frac{F^2\tau^3}{2m} = \tau\Delta E = \frac{3F}{v}A_{\text{inertial, fall}},$$

with $\Delta E = F^2\tau^2/2m$ and the area of N01. The displacement's position component is the sagitta of Lemma X and its momentum component is the accumulated impulse of Proposition I.

Resource bound. The apparatus is said to have **aperture** (L, P) when every state occurring in the experiment satisfies $\Delta_\psi \hat{y} \leq L$ and $\Delta_\psi \hat{p} \leq P$, where $\Delta_\psi X = \|(X - \langle X \rangle_\psi)\psi\|$. Confinement to a region of diameter $2L$ and to momenta of modulus at most P gives this for every state at once. Set $A = LP$; since $\Delta \hat{y} \Delta \hat{p} \geq \hbar/2$ for any state that occurs, $A \geq \hbar/2$.

2. Single shot

Theorem 1. Let the preparation be a pure state ψ with $\Delta_\psi \hat{y} \leq L$ and $\Delta_\psi \hat{p} \leq P$, let the two hypotheses run undisturbed on $[0, \tau]$, and let any measurement on the final state decide between them with equal priors and error probability at most $\epsilon < 1/2$. Then

$$\frac{F\tau L}{\hbar} + \frac{F\tau^2 P}{2m\hbar} \geq \theta(\epsilon) := \arccos\left(2\sqrt{\epsilon(1-\epsilon)}\right), \quad \theta(0) = \frac{\pi}{2}.$$

Proof. The two final states are $U_I\psi$ and $U_F\psi = e^{i\varphi}U_ID(\alpha, \beta)\psi$ by Lemma 1 with $\alpha = \alpha(\tau)$, $\beta = \beta(\tau)$; a common unitary and a global phase change no error probability, so the task is to distinguish ψ from $D(\alpha, \beta)\psi$. Helstrom's bound for two pure states with equal priors gives $\epsilon \geq \frac{1}{2}(1 - \sqrt{1 - |\langle \psi | D | \psi \rangle|^2})$, so $|\langle \psi | D | \psi \rangle| \leq 2\sqrt{\epsilon(1-\epsilon)}$. Write $D = e^{-iG}$ with $G = -(\beta\hat{y} - \alpha\hat{p})/\hbar$. The Mandelstam–Tamm bound in its geometric form states that the Fubini–Study angle between ψ and $e^{-iG}\psi$ grows at rate at most $\Delta_\psi G$, so $|\langle \psi | e^{-iG} | \psi \rangle| \geq \cos(\Delta_\psi G)$ whenever $\Delta_\psi G \leq \pi/2$. Hence either $\Delta_\psi G > \pi/2 \geq \theta(\epsilon)$, or $\cos(\Delta_\psi G) \leq 2\sqrt{\epsilon(1-\epsilon)}$ and again $\Delta_\psi G \geq \theta(\epsilon)$. Finally Δ_ψ is a seminorm on observables, being the norm of the centred operator applied to ψ , so

$$\Delta_\psi G \leq \frac{|\beta|\Delta_\psi \hat{y} + |\alpha|\Delta_\psi \hat{p}}{\hbar} \leq \frac{F\tau L}{\hbar} + \frac{F\tau^2 P}{2m\hbar}. \quad \square$$

The bound uses no measurement model: it holds for every final measurement, optimal or not, and it is a statement about the two states alone.

3. Every protocol of instruments

Theorem 2. Let instruments $M_1, \dots, M_k$ act at times $0 < t_1 < \dots < t_k \leq \tau$, each with rank-one Kraus operators, identical under the two hypotheses, with arbitrary dependence of later instruments on earlier outcomes, followed by any final measurement. Let the apparatus have aperture (L, P) , in the sense that every conditional state arising in the protocol satisfies the two spread bounds. If the full record decides between the hypotheses with equal priors and error probability at most ϵ , then

$$\frac{F\tau L}{\hbar} + \frac{F\tau^2 P}{2m\hbar} \geq 1 - 2\epsilon.$$

Proof. Pass to the interaction picture of U_I , absorbing the free evolution into the instruments, so that under I the state is unchanged between instruments and under F it is acted on, between t_{r-1} and t_r , by $V_r = W(t_r)W(t_{r-1})^\dagger = e^{i\varphi_r}D(\Delta\alpha_r, \Delta\beta_r)$ with $\Delta\alpha_r = \alpha(t_r) - \alpha(t_{r-1})$ and $\Delta\beta_r = \beta(t_r) - \beta(t_{r-1})$, using Lemma 1 and the composition of displacements up to phase; set $t_0 = 0$ and let $r = 1, \dots, k+1$ with $t_{k+1} = \tau$.

Let P_I and P_F be the distributions of the full record. Define hybrid processes $H_0, \dots, H_{k+1}$, where H_s applies V_r for $r \leq s$ and the identity for $r > s$; then $H_0 = I$ and $H_{k+1} = F$. Consecutive hybrids differ by one insertion of V_s into an otherwise identical sequence, so by the triangle inequality for total variation and the data-processing inequality for the common remainder of the protocol,

$$\text{TV}(P_I, P_F) \leq \sum_s \text{TV}(P_{H_{s-1}}, P_{H_s}) \leq \sum_s \sup_{\sigma} \frac{1}{2} \|\sigma - V_s \sigma V_s^\dagger\|_1,$$

the supremum over the conditional states σ reachable at step s . Each such σ is pure, the Kraus operators having rank one, so $\frac{1}{2} \|\sigma - V_s \sigma V_s^\dagger\|_1 = \sqrt{1 - |\langle \sigma | V_s | \sigma \rangle|^2} \leq \sin(\Delta_\sigma G_s) \leq \Delta_\sigma G_s$ when $\Delta_\sigma G_s \leq \pi/2$, by Mandelstam–Tamm as in Theorem 1, and the left side is at most $1 \leq \Delta_\sigma G_s \cdot (2/\pi)^{-1}$ otherwise; in both cases it is at most $\Delta_\sigma G_s$. Therefore

$$\text{TV}(P_I, P_F) \leq \sum_s \frac{|\Delta\beta_s|L + |\Delta\alpha_s|P}{\hbar} = \frac{L \sum_s |\Delta\beta_s| + P \sum_s |\Delta\alpha_s|}{\hbar}.$$

Both $\alpha(t) = -Ft^2/2m$ and $\beta(t) = Ft$ are monotone on $[0, \tau]$, so the increments telescope in absolute value: $\sum_s |\Delta\beta_s| = F\tau$ and $\sum_s |\Delta\alpha_s| = F\tau^2/2m$, **independently of the number of instruments and of their times**. An equal-prior test with error probability at most ϵ has $\text{TV}(P_I, P_F) \geq 1 - 2\epsilon$. $\square$

The monotonicity of α and β is the whole of protocol-universality. Inserting marks subdivides the displacement without enlarging its total variation, so a protocol with a thousand instruments faces the same budget as one with none. This is the probabilistic counterpart of Corollary 4 of the derivation note, and it is the point at which the present framework answers Ozawa: an instrument whose disturbance is correlated with its outcome changes the conditional states σ , and therefore matters only through the apertures L and P that those states respect.

Corollary 3 (the floor). Under Theorem 2, with $A = LP$ fixed and the aperture shape free,

$$\tau\Delta E = \frac{F^2\tau^3}{2m} \geq \frac{(1-2\epsilon)^2\hbar^2}{4A}, \quad \text{and at } A = \frac{\hbar}{2}: \quad \tau\Delta E \geq \frac{(1-2\epsilon)^2\hbar}{2}.$$

Proof. Theorem 2 gives $F \geq 2m\hbar(1-2\epsilon)/[\tau(2mL + \tau P)]$, so $\tau\Delta E \geq 2m\hbar^2(1-2\epsilon)^2\tau/(2mL + \tau P)^2$. At fixed $A = LP$ the denominator $2mL + \tau A/L$ is minimized at $L = \sqrt{\tau A/2m}$ with value $2\sqrt{2m\tau A}$, whose square is $8m\tau A$. $\square$

A larger aperture lowers the floor, and the floor is positive exactly when the aperture is bounded. The maximum of the floor over apertures is attained at the smallest one the uncertainty relation allows, and equals $\hbar/2$ as $\epsilon \rightarrow 0$.

4. The counterexample: universality fails without the aperture

Proposition 4. Fix $m, F, \tau > 0$ and put $\eta = \tau\Delta E/\hbar$. For every $\eta < 1$ there are a pure preparation and a final measurement distinguishing the hypotheses with error probability at most $C\eta$ for an absolute constant C . The preparation has position spread of order $\hbar/(F\tau)$, which diverges as $\eta \rightarrow 0$ at fixed τ and m .

Proof. Write $\alpha = F\tau^2/2m$ and $\beta = F\tau$, so $\alpha\beta = \tau\Delta E = \eta\hbar$. Let χ be a fixed real normalized profile supported in $[-1/2, 1/2]$ with $\|\chi'\|_2 < \infty$, and set

$$\psi = \frac{1}{\sqrt{2}}(\chi_+ + \chi_-), \quad \chi_{\pm}(y) = w^{-1/2}\chi\left(\frac{y \mp d/2}{w}\right), \quad d = \frac{\pi\hbar}{\beta}, \quad w = \sqrt{\frac{\alpha\hbar}{\beta}}.$$

Then $w/d = \sqrt{\alpha\beta/\hbar}/\pi = \sqrt{\eta}/\pi$ and $\alpha/w = \sqrt{\alpha\beta/\hbar} = \sqrt{\eta}$, so for $\eta < 1$ the two packets are disjoint and the displacement is small compared with the packet width. Up to a global phase, $(D(\alpha, \beta)\psi)(y) = e^{i\beta(y-\alpha/2)/\hbar}\psi(y-\alpha)$. Since the supports of χ_+ and χ_- are separated by $d - w > d/2$ while the shift is $\alpha < w$, the cross terms vanish and

$$\langle \psi | D | \psi \rangle = \frac{1}{2} \sum_{\pm} e^{\pm i\beta d/(2\hbar)} \int \overline{\chi_{\pm}(y)} e^{i\beta(y \mp d/2)/\hbar} \chi_{\pm}(y - \alpha) dy.$$

With $d = \pi\hbar/\beta$ the two prefactors are $e^{\pm i\pi/2}$ and cancel the leading terms, leaving the two residuals. Each residual integral differs from 1 by at most $\|\chi(\cdot) - \chi(\cdot - \alpha/w)\|_2 + \beta w/\hbar \leq (\alpha/w)\|\chi'\|_2 + \beta w/\hbar$, so

$$|\langle\psi|D|\psi\rangle| \leq \frac{\alpha}{w}\|\chi'\|_2 + \frac{\beta w}{\hbar} = \sqrt{\eta}(\|\chi'\|_2 + 1).$$

Helstrom's optimal test then has error probability $\frac{1}{2}(1 - \sqrt{1 - |\langle\psi|D|\psi\rangle|^2}) \leq \frac{1}{4}|\langle\psi|D|\psi\rangle|^2 \leq C\eta$ with $C = (\|\chi'\|_2 + 1)^2/4$. The position spread of ψ is $d/2 + O(w) = \pi\hbar/(2F\tau) + O(w)$. $\square$

The counterexample saturates Theorem 2. Its aperture has $L \simeq \pi\hbar/(2F\tau)$, for which the first term of the theorem's left side is about $\pi/2$, so the necessary condition is met with no room to spare. The momentum spread is of order $\hbar/w = \sqrt{2m\hbar/\tau}$, which stays bounded as $\eta \rightarrow 0$; the diverging resource is the separation, not the energy. So the obstruction to reading Newton's shrinking sagitta is a bound on **how large the apparatus may be**, and a laboratory of unbounded extent has no Planck gap in this comparison.

5. Position against the standard quantum limit

The combination $F^2\tau^3 \gtrsim m\hbar$ is the standard quantum limit for detecting a force on a free mass (Braginsky and Khalili, *Quantum Measurement*, 1992, metadata). Its status has been disputed since Yuen's objection: Caves defended it (PRL **54**, 2465, 1985, abstract) and Ozawa exhibited a measurement breaking it for free-mass position (PRL **60**, 385, 1988, abstract). The present statement is not a new bound of that kind and does not adjudicate that dispute on its own terms. Three differences fix its place.

- **Universality over protocols is proved, not assumed.** The usual derivations fix a monitoring scheme and balance its imprecision against its back-action. Theorem 2 quantifies over all finite sequences of instruments with arbitrary adaptivity, and the reason it can is structural: the signal enters as a phase-space path of bounded total variation, and subdividing a path does not lengthen it.
- **The resource is named and the bound is tight in it.** Ozawa's construction and Proposition 4 both buy distinguishability with preparations of large phase-space extent. Corollary 3 prices that exchange as $\tau\Delta E \geq (1 - 2\epsilon)^2\hbar^2/(4A)$, and Proposition 4 shows the price is right, so the standard quantum limit appears here as the special case $A = \hbar/2$ rather than as a law.
- **The criterion is hypothesis testing, not estimation.** The quantity bounded is the error probability of deciding between two known histories, which is the question Newton's geometry asks, and not the variance of an estimate of F .

Relative to the error-disturbance literature, the argument uses no such relation. It uses only Mandelstam–Tamm and the uncertainty relation through $A \geq \hbar/2$, which is why the contested calibration questions of Ozawa and of Busch, Lahti and Werner (PRL **111**, 160405, 2013, abstract) do not enter.

6. What this settles for the programme

Against the classical theorem of the derivation note the parallel is exact in form and opposite in direction. Classically the floor of the recorded comparison is $\tau\Delta E \geq \frac{9}{2}\kappa$, with $\kappa = \delta\Delta$ the phase-space cost of a single mark, so **coarser marks raise the floor**. Quantum mechanically the floor is $\tau\Delta E \geq (1 - 2\epsilon)^2\hbar^2/(4A)$, with A the phase-space aperture of the apparatus, so **a larger laboratory lowers it**. The two meet where the aperture is as small as the uncertainty relation permits: there $A = \hbar/2$ and the floor is $\hbar/2$, which is the order of the value $\kappa = \Lambda p = \hbar/2$ that Newton's interval of fits and corpuscle impulse supply in the mark-floor note.

The Newton-age reading is therefore sharper than before. Newton's limit in Lemmas X and XI takes the sagitta and the swept area to zero, and nothing in the geometry stops it. What stops the *recorded* comparison is the phase-space area of the apparatus, and the two Newtonian quantities that enter are precisely the two components of the displacement: the sagitta $F\tau^2/2m$ against the momentum spread, and the impulse $F\tau$ of Proposition I against the position spread. Newton had both quantities and had no reason to pair them with spreads of anything.

7. Consequence for STATE

The goal of proving the probabilistic form is discharged. Theorem 1 gives the single-shot bound with the sharp constant $\arccos(2\sqrt{\epsilon(1-\epsilon)})$, Theorem 2 extends it to every finite adaptive protocol of rank-one instruments with the constant $1 - 2\epsilon$, Corollary 3 converts it to the floor $\tau\Delta E \geq (1 - 2\epsilon)^2\hbar^2/(4A)$, and Proposition 4 decides the universality question: universality over instruments holds, and universality over preparations fails, with an explicit family that saturates the theorem. Positioning against the standard quantum limit is in Section 5, and no error-disturbance relation is used.

Open, in order:

1. **Mixed conditional states.** Theorem 2 assumes rank-one Kraus operators so that every conditional state is pure. The extension needs a bound on $\|\sigma - V\sigma V^\dagger\|_1$ for mixed σ in terms of a spread, where the natural route is a purification with the aperture imposed on the purified state.
2. **The sharp constant.** Theorem 1 gives $\theta(\epsilon)$ and Theorem 2 gives $1 - 2\epsilon$ for the same quantity; the gap between them is an artifact of the telescoping step, and closing it would also close the interval $[9, 36]$ left by the classical theorem.
3. **The general force law.** Both theorems use only that the phase-space path $t \mapsto (\alpha(t), \beta(t))$ has total variation $F\tau^2/2m$ and $F\tau$ in its two components. For a general force the statement is that the two total variations, weighted by P and L , must exceed $\hbar(1 - 2\epsilon)$; the constant-force case is then one evaluation.
4. **Publication.** With Section 5 in place the foundations paper has its positioning; what remains before submission is item 1, item 2 and a literature pass on resource-bounded quantum metrology, where a bound of the shape of Corollary 3 may already exist for estimation rather than testing.